

Postoperative Pain Outcomes Following Root Canal Treatment: Electronic Apex Locators versus Radiographic Working Length Determination

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ABSTRACT

One hundred twenty-five patients with asymptomatic irreversible pulpitis underwent single-visit root canal treatment with working length determined via one of five methods. Postoperative pain assessments revealed that discomfort gradually diminished over the 48-hour evaluation window in all treatment groups. Statistical analysis comparing pain intensities across the five working length determination approaches showed no clinically or statistically significant differences at any assessment timepoint. Pain patterns were similar whether patients underwent electronic locator-guided or radiography-based working length determination. Variables including age, sex, tooth location, and analgesic intake patterns did not demonstrate significant associations with postoperative pain across groups. Medication use for post-treatment discomfort management showed no meaningful variation among the five treatment approaches. These findings indicate that under standardised clinical conditions, the specific working length determination methodology does not substantially influence pain experience following root canal completion in this patient population with straightforward single-rooted anatomy.

Trial Context

Underlying randomised trial, five treatment arms comparing periapical radiography for working length determination with various electronic apex locators for working length determination.